

Trainings ▾

Preparatory Steps

- Remove the turbocharger. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/89/89-010-033.html>) in Section 10.
- Remove the exhaust connection pipe (power generation **only**). Refer to Procedure 011-001 (</qs3/pubsys2/xml/en/procedures/89/89-011-001.html>).

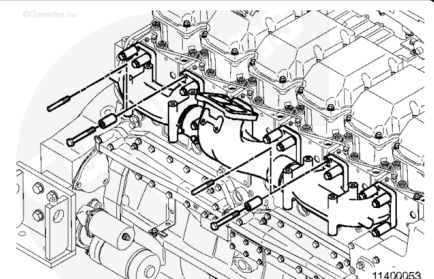


Remove

Use two guide studs.

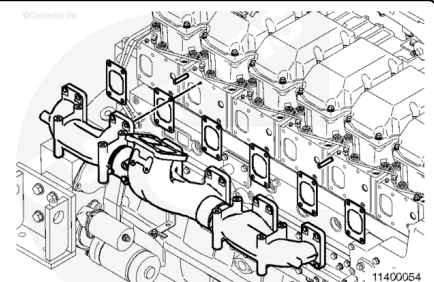
Remove the two capscrews, as shown.

Install the guide studs, as shown.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



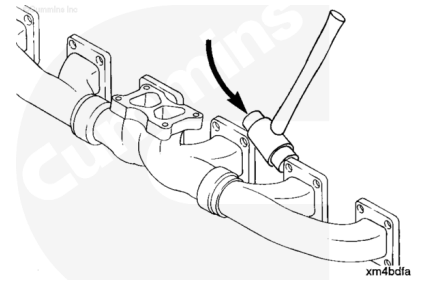
Remove the remaining capscrews.

Remove the exhaust manifold and the gaskets.

Discard the gaskets.

Disassemble

Use a mallet to remove the end sections from the exhaust manifold center section.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

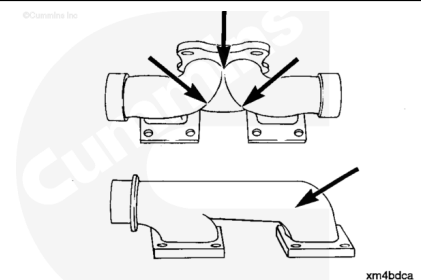
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ CAUTION ⚠

Care must be taken to not destroy the exhaust manifold heat shield insulation while cleaning.

Use steam or solvent to clean the exhaust manifold. Use a wire brush to clean the sealing joint inside and outside diameters, removing any scale that has accumulated.

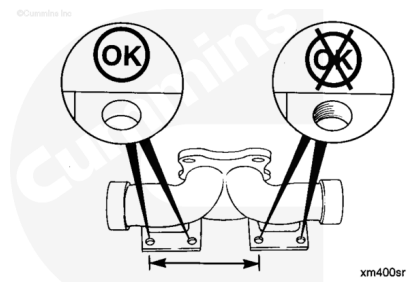
Inspect for cracks in the areas shown.



Inspect the capscrew holes in the center section for damage.

The manifold has been subjected to high temperatures if the capscrew threads are visible on the inside of the capscrew holes. High temperature can cause the manifold to shrink.

Measure the center-to-center distance between the same position capscrew holes of the two flanges on the center section and both end sections to determine if the manifold capscrew mounting holes are out of location.

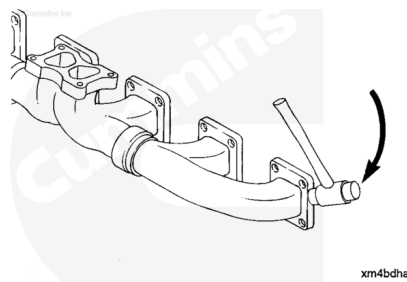


Exhaust Manifold Capscrew Hole Location

mm		in
213	MIN	8.39
215	MAX	8.46

Assemble

Use a mallet to drive the end section into the center section.



Install

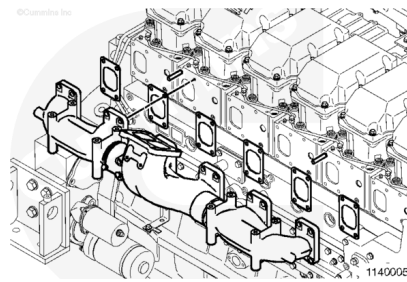
⚠ CAUTION ⚠

Do not use gasket cement or the gasket will fail.

Install the guide studs used during removal.

Use contact adhesive to attach the gasket on the cylinder head exhaust ports.

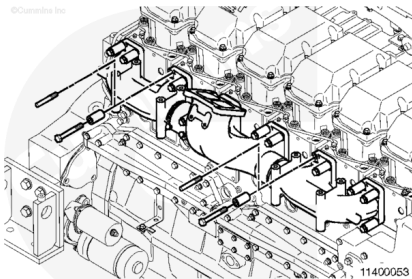
Install the exhaust manifold.



Apply an antiseize compound on the capscrew threads.

Install the capscrews.

Remove the two guide studs and install the remaining capscrews.



Tighten the capscrews in the sequence shown.

Torque Value:

Step 1

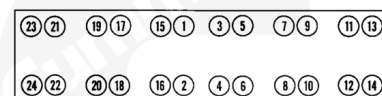
1. 55 n•m [40 ft-lb]

Torque Value:

Step 2

1. 120 n•m [89 ft-lb]

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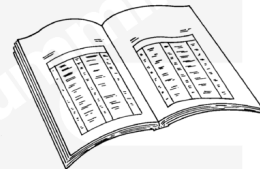


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Finishing Steps

- Install the exhaust connection pipe (power generation **only**). Refer to Procedure 011-001 (</qs3/pubsys2/xml/en/procedures/89/89-011-001.html>) in Section 11.
- Install the turbocharger. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/89/89-010-033.html>) in Section 10.

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